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SIC-AI Capstone Project Proposal

Speaker-Aware Noise Cancellation and Target Speech Enhancement

Submitted By

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Abstract

This project develops a speech enhancement system that uses a short enrollment recording of a target speaker to extract and enhance the target speaker's voice from a single-channel microphone mixture containing background noise and competing speakers, while suppressing unwanted noise and interfering speech.

The system combines three components: (i) a pretrained ECAPA-TDNN speaker encoder that maps a 3–5 second enrollment clip to a 192-dimensional speaker embedding; (ii) a causal target-speaker extraction network (based on Conv-TasNet and BSRNN backbones) that operates on 16 kHz STFT features and estimates a target-speaker mask using only past and current frames; and (iii) a streaming audio pipeline that applies the mask, reconstructs the waveform via inverse STFT, and outputs the enhanced speech. Training uses synthetic mixtures of clean speech, interfering speakers, and environmental noise at controlled SNRs and target-to-interferer ratios, with curriculum learning from easy to hard conditions and augmentation for reverberation and channel variability.

We evaluate the system on speaker-disjoint and noise-disjoint test sets using SI-SDR, STOI, and PESQ for speech quality and separation, and measure model size and computational cost on a consumer CPU. The final deliverable is a CPU-deployable model that demonstrates selective preservation of an unseen target speaker in realistic meeting scenarios, together with a reproducible training and evaluation framework.

Keywords: Target Speaker Extraction, Speech Separation, ECAPA-TDNN, Conv-TasNet, BSRNN, Causal Neural Networks, Speaker Embedding, STFT

1. Introduction

1.1 Background

Voice communication systems capture not only the intended speaker's speech but also surrounding acoustic events: fan or air-conditioner hum, keyboard sounds, traffic, nearby conversations, and other transient sounds. This interference reduces intelligibility and makes communication difficult, especially in shared or noisy environments.

Speech enhancement addresses the problem of recovering a cleaner speech signal from a noisy recording. Neural speech enhancement systems have shown that learned time-frequency representations can outperform traditional signal-processing approaches under challenging conditions, and the Deep Noise Suppression (DNS) Challenge established a widely used benchmark for single-channel enhancement [1]. Lightweight architectures such as DeepFilterNet have further demonstrated that competitive quality is achievable under strict computational budgets [2].

However, conventional noise suppression does not distinguish between the desired speaker and another human speaker in the same recording. In a meeting environment, preserving *all* speech is often insufficient – a second person talking nearby, a television in the background, or a roommate's conversation is not noise in the traditional sense but is still undesired. Target speaker extraction addresses this more specific problem: given a mixture and a short reference recording of the desired speaker, the system extracts that speaker's voice while suppressing everything else, including other speakers. Recent work on target-speaker extraction explicitly studies causal models that can operate under latency constraints similar to those targeted here [11, 12].

1.2 Problem Statement

Existing microphone noise-suppression tools reduce environmental noise reasonably well but do not handle the case where multiple people are speaking near the same microphone. A conventional denoiser preserves both speakers because both are classified as speech, which is not always the desired behaviour in a one-on-one or one-to-many online call.

> *How can a compact, causal, speaker-conditioned neural network be designed and trained to isolate an unseen target speaker from realistic mixtures containing competing speech and environmental noise, while satisfying low end-to-end latency and CPU-only constraints?*

1.3 Objectives

The project has three main objectives:

1. **Design and implement a target-speaker extraction system.** Integrate a pretrained ECAPA-TDNN speaker encoder with a causal separator (Conv-TasNet/BSRNN) to extract a target speaker from mixtures containing competing speech and environmental noise, operating fully causally.
2. **Evaluate quality, robustness, and efficiency.** Benchmark the system on speech quality (SI-SDR, STOI, PESQ), target-speaker isolation, and generalization to unseen speakers and noise, while measuring computational cost and model size.
3. **Deliver a reproducible training and evaluation framework.** Provide a complete pipeline for dataset preparation, synthetic mixture generation, model training, validation, and performance evaluation.

2. Literature Review

Deep Noise Suppression (DNS) Challenge

The INTERSPEECH DNS Challenge established a standard dataset, evaluation protocol, and both objective and subjective testing framework for single-channel speech enhancement, and has become a common reference point for comparing denoising systems [1].

DeepFilterNet

DeepFilterNet is a low-complexity speech enhancement framework built around deep filtering in the time-frequency domain, designed specifically for full-band audio under tight computational constraints [2]. It is a useful reference for what a lightweight, causal, CPU-deployable denoiser can achieve.

VoiceBank+DEMAND

This is a widely used benchmark pairing clean speech with environmental noise at controlled SNRs, and is a standard training/evaluation set for supervised speech enhancement [3]. It will serve as one of the primary sources of clean speech and noise mixtures for training and benchmarking.

Conv-TasNet and Time-Domain Separation

Conv-TasNet reformulated speech separation as a time-domain, learned-basis problem and showed that time-domain masking can surpass ideal time-frequency magnitude masking on the source-separation task [4]. The causal variant replaces global layer normalization (gLN) with cumulative layer normalization (cLN) and uses only left-padded convolutions to maintain causality, at a modest cost in SI-SDR [4]. Public implementations document hyperparameters such as encoder filters ($N = 512$), kernel length ($L = 16$), bottleneck channels ($B = 128$), hidden channels ($H = 512$), kernel size ($P = 3$), blocks per repeat ($X = 8$), and repeats ($R = 3$), which provide a concrete starting point [4, 5].

ECAPA-TDNN Speaker Embeddings

ECAPA-TDNN is a state-of-the-art speaker verification architecture that combines SE-Res2Net blocks, multi-layer feature aggregation, attentive statistics pooling, and additive angular margin (AAM) softmax [6]. It produces compact 192-dimensional embeddings that are robust across acoustic conditions and has become a standard choice for speaker-conditioned tasks, including target-speaker extraction [6, 7, 8]. Recent TSE work explicitly uses ECAPA-TDNN embeddings to condition separation networks [9].

Target Speaker Extraction

Speaker-conditioned extraction – using a short enrollment utterance to bias a separation network toward a specific voice – is a well-studied variant of speech separation. VoiceFilter introduced a speaker-dependent "voice filtering" formulation using a reference signal [10]. More recent work studies lightweight Conv-TasNet variants and state-space models, reporting computational efficiency on consumer CPUs [11]. TargetVoice reports low processing latency and 8–10 dB SI-SDR improvements under challenging signal-to-interference ratios [12]. Other systems report CPU RTFs around 0.66–0.94 for target sound extraction [13].

Conditioning Mechanisms

FiLM (feature-wise linear modulation) has been used to condition separation networks on speaker embeddings, often outperforming simple concatenation [14, 15]. Multiplicative gating and other embedding-driven modulations are also explored in speaker-aware separation [16]. Ablating concatenation, FiLM, and gating is a standard way to add rigor to TSE papers.

3. Methodology

3.1 System Overview

The system operates in a single mode: target-speaker extraction. Given a mixture $X = S_t + S_i + N$ and a short enrollment recording R_t of the target speaker, the model estimates a target-speaker mask M_t and reconstructs $\hat{S}_t = M_t \odot X$ via inverse STFT. The architecture consists of three main components:

- A pretrained ECAPA-TDNN speaker encoder producing a 192-dimensional embedding E_t.
- A causal separator backbone (Conv-TasNet, BSRNN, or TF-GridNet) operating on STFT features.
- A conditioning module that fuses E_t with the separator's internal representations.

3.2 Audio Acquisition and Time-Frequency Representation

Mono microphone audio is captured at 16 kHz and divided into short overlapping frames for streaming processing. The short-time Fourier transform converts each frame into a time-frequency representation:

$$X(k, m) = \sum_{n=0}^{N-1} x[n + mH]\, w[n]\, e^{-j2\pi kn/N},$$

where N is the FFT size, H is the hop size, and $w[n]$ is the analysis window. An initial configuration of a 512-point FFT with a 160-sample hop (32 ms window, 10 ms hop) at 16 kHz will be used as a starting point, subject to latency/quality trade-off experiments. This yields 257 frequency bins for a real-valued signal.

3.3 High-Level Architecture

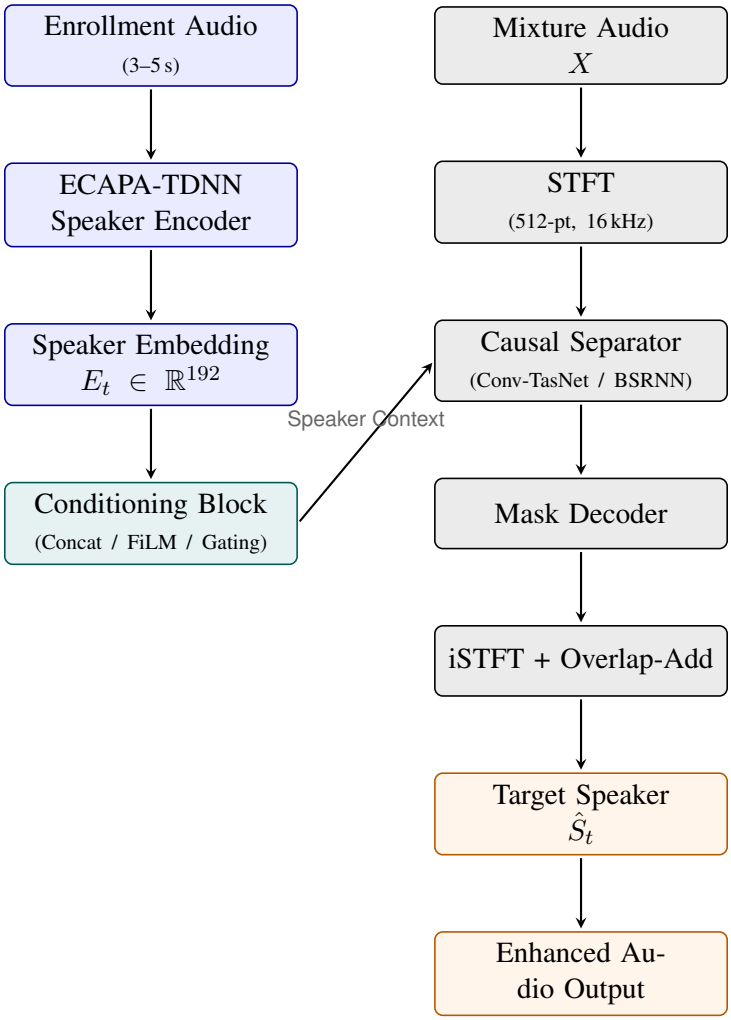


Figure 3.1: Complete target speaker extraction system architecture, detailing enrollment embedding extraction, spectral feature processing, and conditioning integration.

3.4 Speaker Encoder: ECAPA-TDNN

The speaker encoder is a pretrained ECAPA-TDNN model, initialized from a public checkpoint (e.g., SpeechBrain's `spkrec-ecapa-voxceleb`) [7]. Key architectural elements include:

- **Input features:** 80-dim log-Mel filterbank, 25 ms window, 10 ms hop, 16 kHz audio.
- **Backbone:** SE-Res2Blocks (Squeeze-Excitation + Res2Net), typically 3 blocks.
- **Channels:** 512 (lightweight) or 1024 (standard).
- **Kernel/dilation per block:** $k = 3$, dilations $\{2, 3, 4\}$ across the three blocks.

- **Res2Net scale:** $s = 8$.
- **Aggregation:** Multi-layer feature aggregation (concatenation of all SE-Res2Block outputs).
- **Pooling:** Attentive statistics pooling (mean + std, attention-weighted over time).
- **Embedding dim:** 192 (standard).
- **Loss (pretraining):** AAM-Softmax, margin 0.2, scale 30, trained on VoxCeleb2 (6000 speakers).

Two operating modes for the encoder will be compared as an ablation:

- **Frozen encoder:** ECAPA-TDNN weights fixed, only the separator trained.
- **Fine-tuned encoder:** Last feed-forward layer of ECAPA-TDNN retrained jointly with the separator to shift embedding focus toward extraction-relevant information [9].

Separator Backbones

Figure 3.2 shows the internal architecture of the causal separator with conditioning.

Three separator backbones are implemented to span the quality–latency–compute space:

Causal Conv-TasNet (Baseline)

Conv-TasNet is the right starting baseline because its hyperparameters are exactly documented and it is cheap enough to finish training on a student budget. The causal variant uses cumulative layer normalization (cLN) instead of global layer normalization (gLN) and left-padded convolutions only [4].

The causal constraint typically costs a few dB of SI-SDR versus the non-causal version; this gap will be reported explicitly.

Causal BSRNN (Stretch Goal)

Band-split RNN (BSRNN) is a recent separator backbone that achieves strong performance with manageable compute. Recent work reports 32 ms theoretical latency in causal mode and is currently a strong TSE separator backbone [17]. BSRNN will be implemented in causal mode as a stronger baseline, with quality compared against Conv-TasNet.

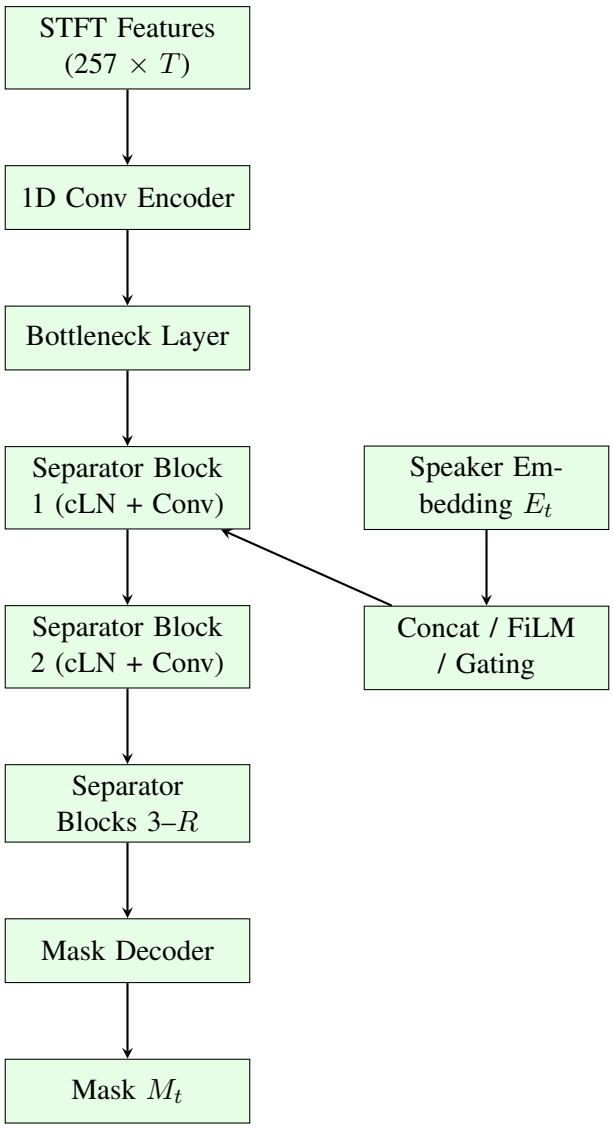


Figure 3.2: Causal separator architecture with speaker conditioning. Each separator block uses cumulative layer normalization (cLN) and left-padded convolutions.

TF-GridNet (Upper Bound)

TF-GridNet is a higher-quality, heavier-compute separator that will be used as a non-causal upper-bound reference even if it is not made causal [18]. It provides a target for what is achievable in terms of SI-SDR and STOI when latency is not constrained.

3.5 Conditioning Mechanisms

The speaker embedding $E_t \in \mathbb{R}^{192}$ is fused with the separator's acoustic features via one of three mechanisms, evaluated as an ablation:

- **Concatenation (simplest):** Tile E_t across time and concatenate to bottleneck features.

Baseline.

- **FiLM (feature-wise linear modulation):** E_t is projected to learned scale γ and shift β applied per-channel: $F' = \gamma(E_t) \odot F + \beta(E_t)$. Usually outperforms plain concat [14, 15].
- **Multiplicative gating:** E_t is projected and multiplied elementwise into intermediate features.

SI-SDR will be reported for all three; this three-way ablation is a cheap, standard way to add rigor to the paper.

3.6 Training Data and Augmentation

Training mixtures are synthesized as $X = S_t + \beta S_i + \alpha N$, with mixing coefficients chosen to cover a range of SNRs and target-to-interferer ratios. Noise sources will span fan/AC noise, traffic, keyboard typing, crowd noise, construction, and household sounds, drawn from public datasets (VoiceBank, DEMAND, MUSAN, DNS noise) and a small self-recorded local set used primarily for *unseen*-condition evaluation.

Augmentation includes:

- Random SNR (e.g., -5 to 20 dB) and target-to-interferer ratio.
- Random gain and temporal cropping.
- Noise-segment randomization.
- Limited reverberation simulation using room impulse responses.
- Randomized speaker/noise pairing.

3.7 Loss Functions

Training combines a waveform-level objective with optional spectral regularization:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{SI-SDR}} + \lambda_2 \mathcal{L}_{\text{spec}} + \lambda_3 \mathcal{L}_{\text{mask}},$$

with weights selected via validation experiments. The primary loss is scale-invariant SDR (SI-SDR), maximized [4, 13].

3.8 Causal Inference

The deployed inference path uses only current and past frames – no bidirectional recurrence – and processes chunks continuously. Latency components include:

- STFT window and hop delay.
- Model processing time.
- Overlap-add and iSTFT reconstruction.

The enhanced audio is output as a waveform for offline evaluation and potential integration into communication applications.

4. Evaluation Plan

Quality and Intelligibility Metrics

- **SI-SDR:** Scale-invariant signal-to-distortion ratio improvement.
- **STOI:** Short-time objective intelligibility (0–1 scale).
- **PESQ:** Perceptual evaluation of speech quality.
- **Target-to-interference ratio:** Improvement in target speaker dominance over competing speech.

Computational Metrics

- **End-to-end latency:** Buffering + processing + output.
- **Processing time per second of audio.**
- **CPU utilization:** Average and peak CPU percentage during continuous processing.
- **Memory usage:** Peak RAM and model memory footprint.
- **Parameter count:** Total trainable parameters.

Generalization Tests

- **Seen speakers, seen noise:** Baseline performance.
- **Unseen speakers, seen noise:** Speaker generalization.
- **Seen speakers, unseen noise:** Noise generalization.
- **Unseen speakers, unseen noise:** Combined generalization.
- **Local real-world recordings:** Final robustness evaluation.

5. Expected Outcomes

By the end of the project, the team expects to deliver:

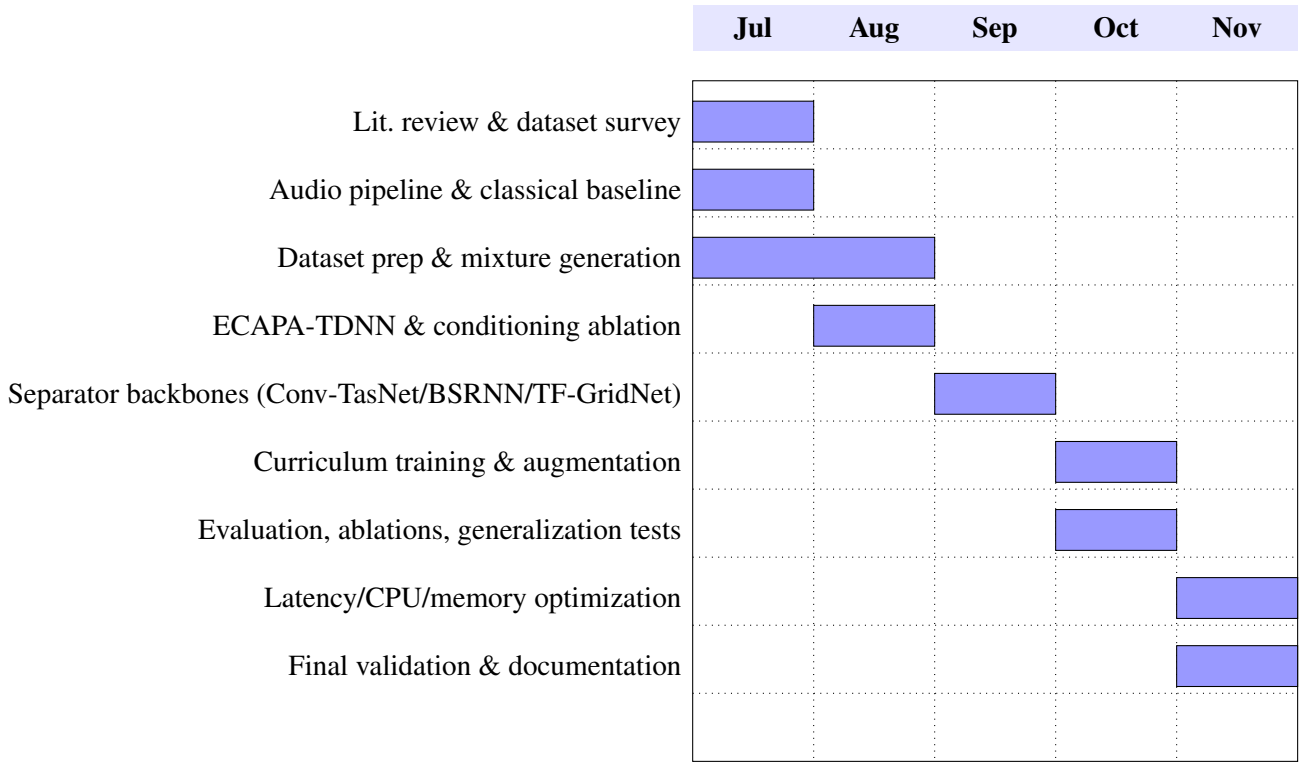
1. A lightweight, trained speaker-independent speech enhancement model capable of reducing background noise while preserving speech quality and intelligibility.
2. A comparative evaluation of different model architectures and speaker-conditioning mechanisms using objective metrics such as SI-SDR, STOI, PESQ, and computational cost, including testing on unseen speakers and unseen noise conditions.
3. A reproducible end-to-end training and evaluation pipeline that covers dataset preparation, synthetic mixture generation, model training, validation, and performance evaluation.

6. Role of Each Team Member

Member	Primary Responsibilities
Austina Aryal	Dataset preparation, speech and noise data collection, preprocessing, and augmentation.
Prashamsa Aryal	Speaker encoder development, target-speaker extraction model, and speaker-conditioning methods.
Sostika Shrestha	Training pipeline, curriculum learning, model training, and inference pipeline.

Note: All three members will jointly contribute to literature review, evaluation-metric implementation, and the final report; the split above reflects primary ownership of each module rather than exclusive responsibility.

7. Timeline


Jul
Aug
Sep
Oct
Nov
Lit. review & dataset survey
Audio pipeline & classical baseline
Dataset prep & mixture generation
ECAPA-TDNN & conditioning ablation
Separator backbones (Conv-TasNet/BSRNN/TF-GridNet)
Curriculum training & augmentation
Evaluation, ablations, generalization tests
Latency/CPU/memory optimization
Final validation & documentation

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